

Supplemental Information

Integrated Genomic Analysis Identifies Clinically Relevant Subtypes of Glioblastoma Characterized by Abnormalities in *PDGFRA*, *IDH1*, *EGFR*, and *NF1*

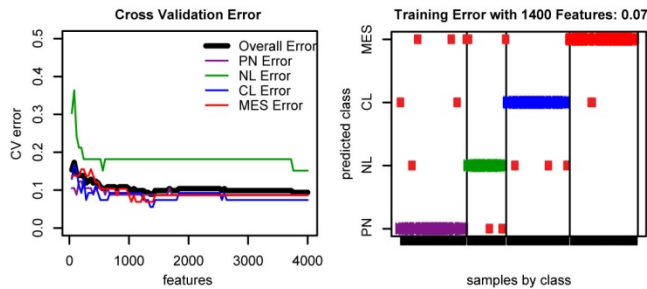
Roel G.W. Verhaak, Katherine A. Hoadley, Elizabeth Purdom, Victoria Wang, Yuan Qi, Matthew D. Wilkerson, C. Ryan Miller, Li Ding, Todd Golub, Jill P. Mesirov, Gabriele Alexe, Michael Lawrence, Michael O'Kelly, Pablo Tamayo, Barbara A. Weir, Stacey Gabriel, Wendy Winckler, Supriya Gupta, Lakshmi Jakkula, Heidi S. Feiler, J. Graeme Hodgson, C. David James, Jann N. Sarkaria, Cameron Brennan, Ari Kahn, Paul T. Spellman, Richard K. Wilson, Terence P. Speed, Joe W. Gray, Matthew Meyerson, Gad Getz, Charles M. Perou, D. Neil Hayes, and The Cancer Genome Atlas Research Network

Inventory

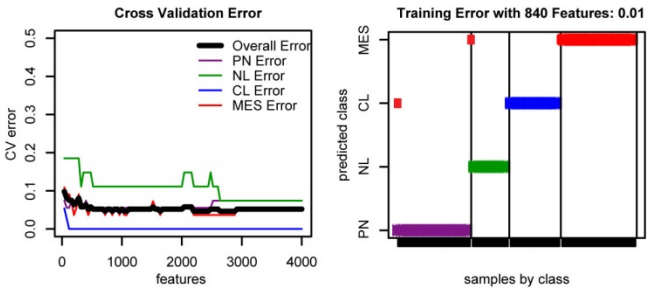
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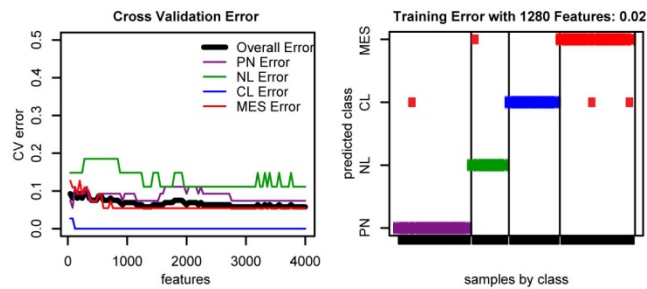
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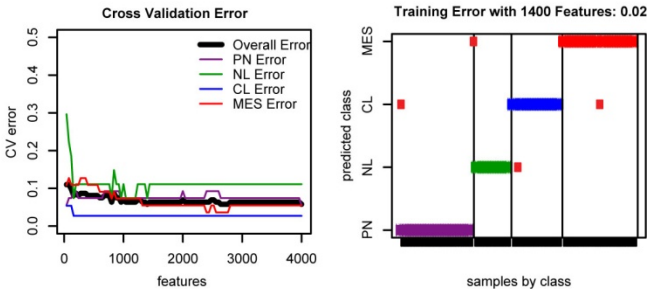
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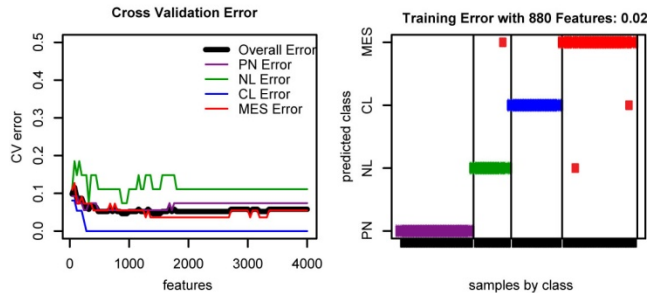
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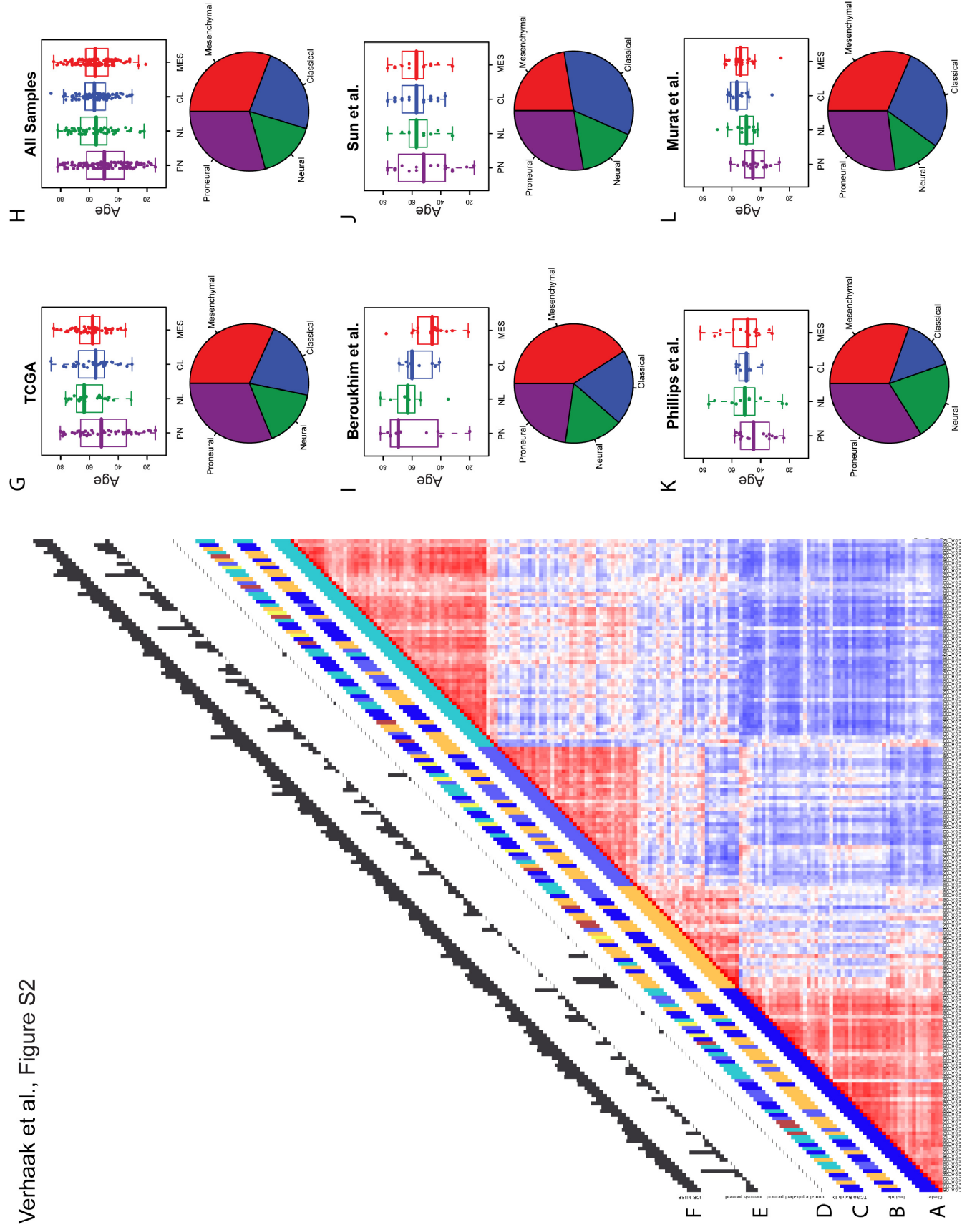
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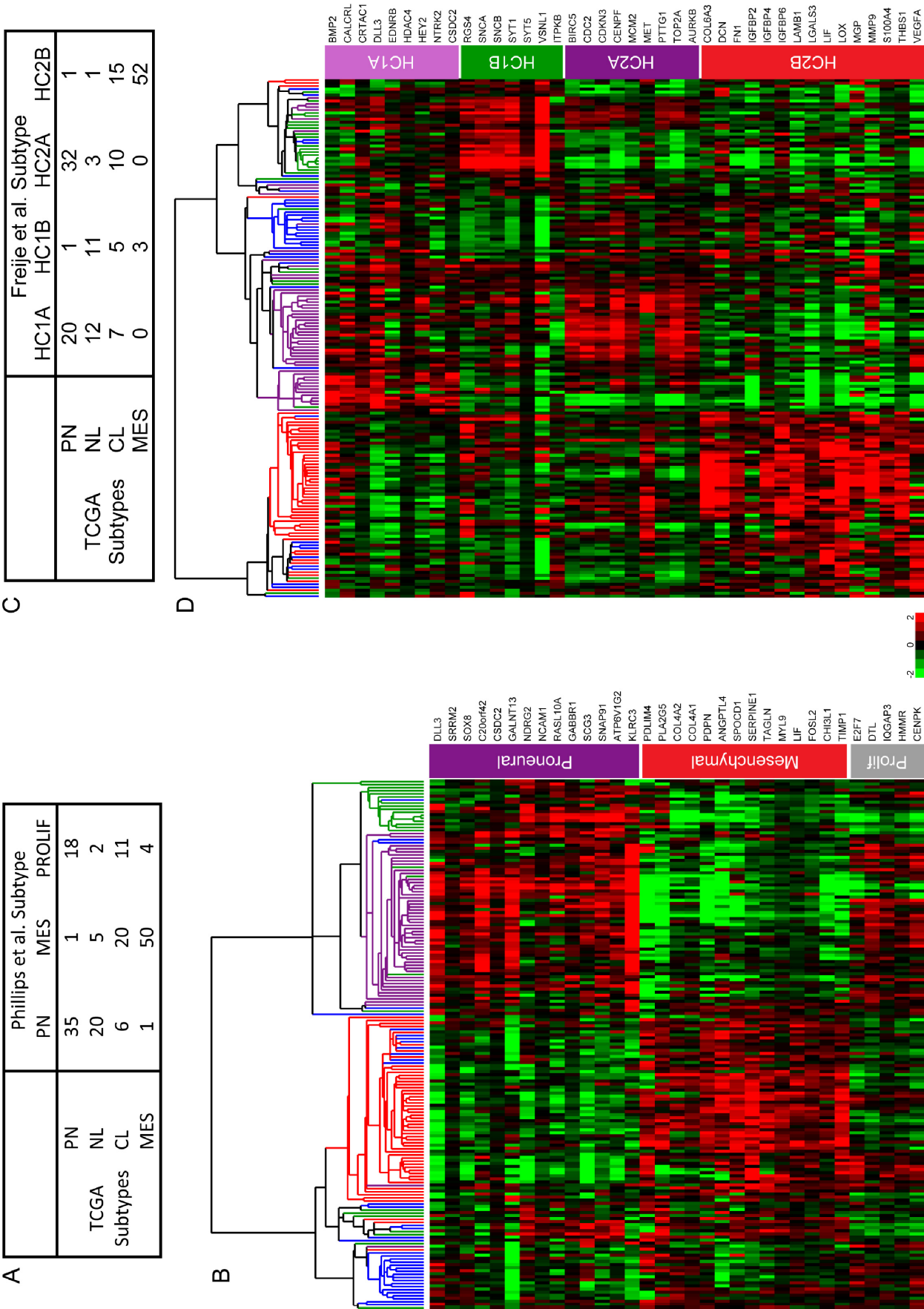
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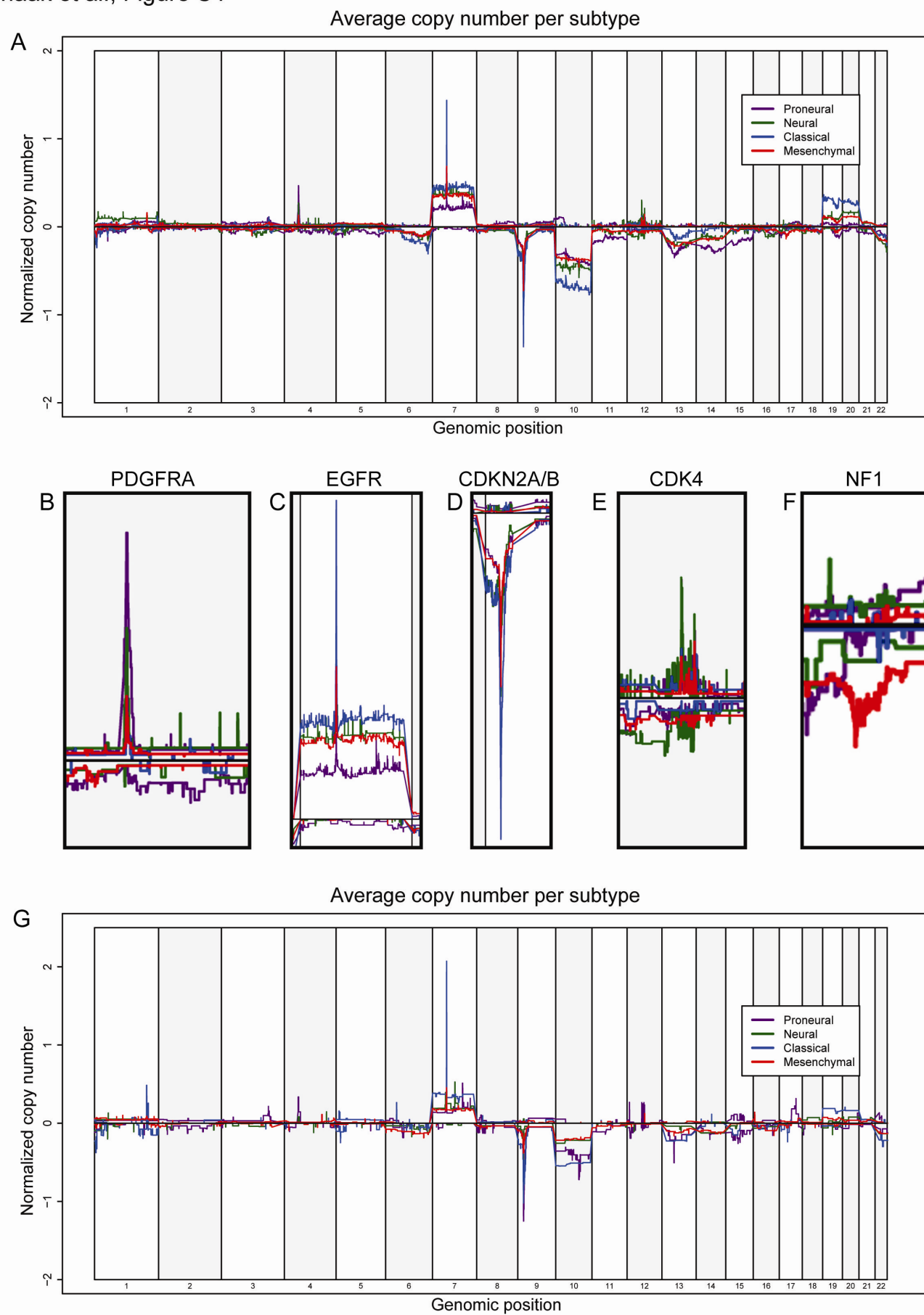


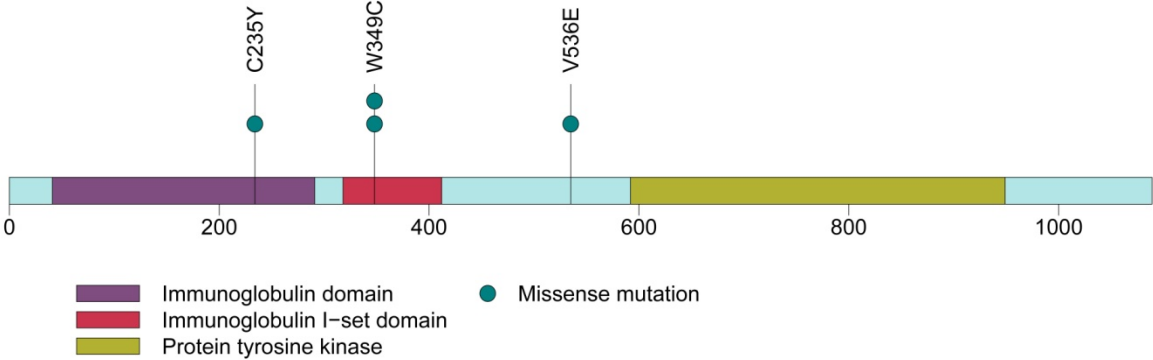
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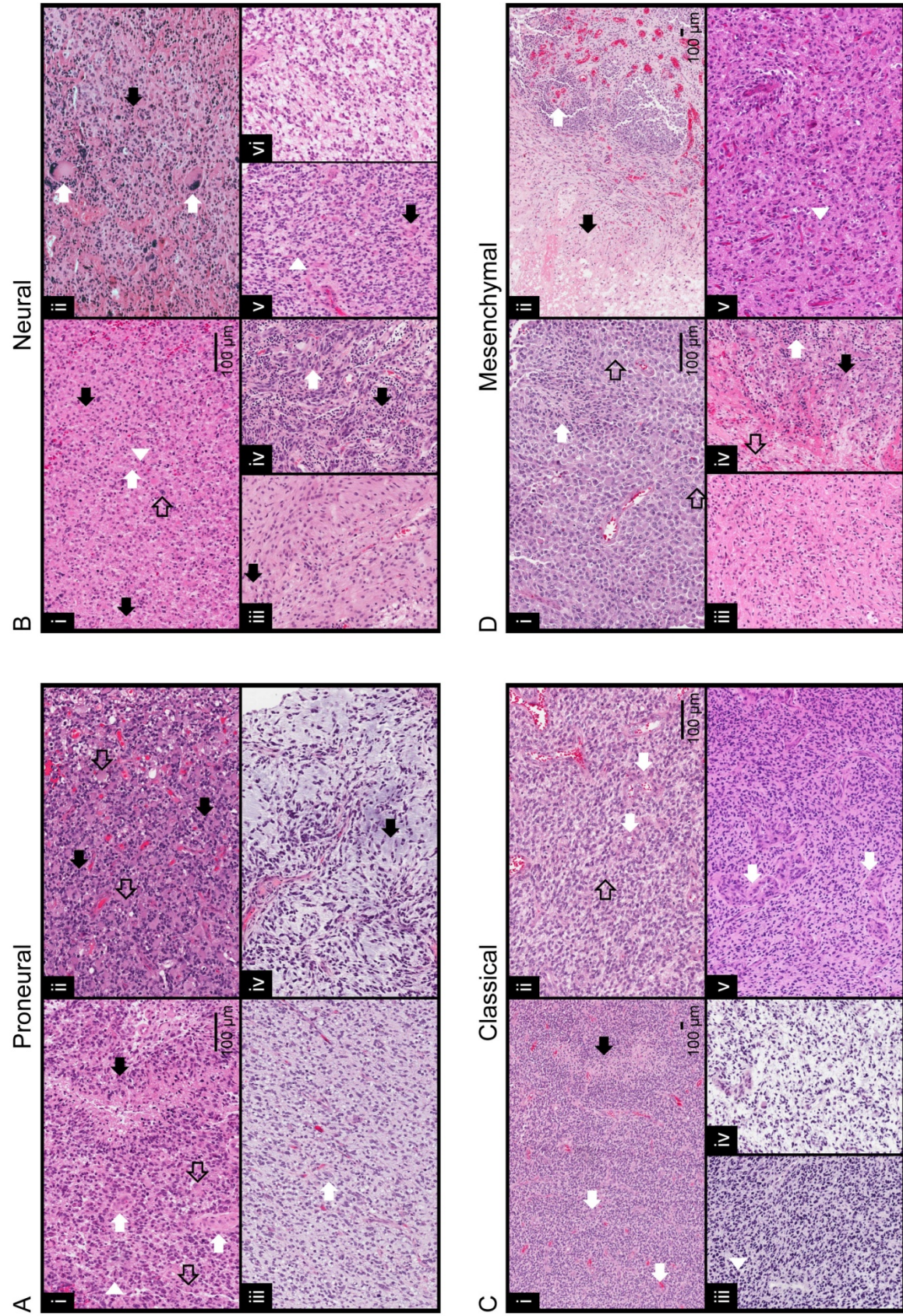
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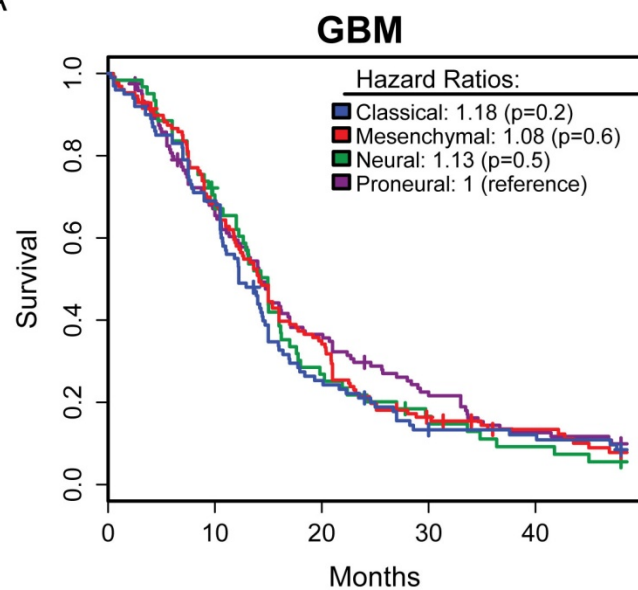




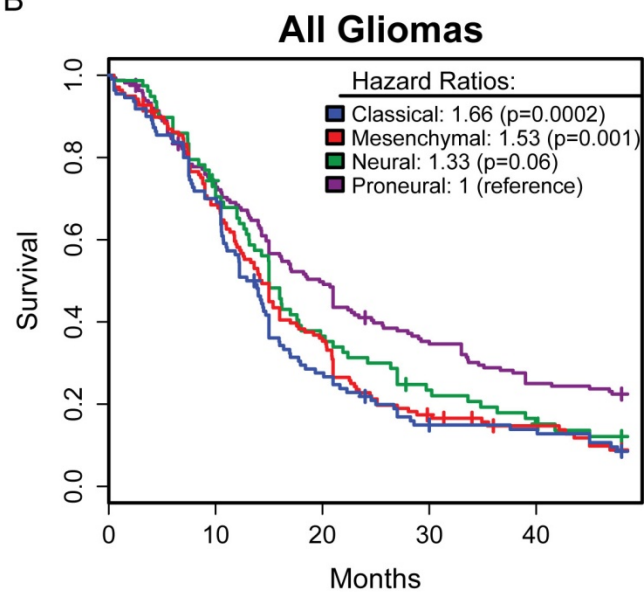
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A



B



Supplemental Figure Legends

Figure S1. Improved cross validation and training error rates using unified core dataset versus each individual dataset. Cross validation and training error rates for (A) Entire unified dataset (n=202), (C) Core unified dataset (n=173), (C) Core Affymetrix U133A, (D) Core Affymetrix-HuEx, and (E) Core Agilent 244K Custom.

Figure S2. (A-F) Sample-sample heatmap showing pair-wise correlation between all samples. The colors of the cells relate to Pearson's correlation coefficient values, with deeper colors indicating higher positive (red) or negative (blue) correlations. Additional information is shown alongside the diagonal. (A) Sample subtype; Proneural in blue, Neural in orange, Classical in purple, Mesenchymal in cyan. (B) Sample providing institute; Henry Ford Hospital in blue, MD Anderson Cancer Center in orange, University of California San Francisco in purple, Duke University in cyan. (C) Batch; batch 1 in purple, batch 2 in brown, batch 3 in yellow, batch 4 in purple, batch 5 in blue, batch 6 in turquoise. (D) The percentage normal tissue in the sample, scaled to a maximum of 100%. (E) Percentage necrosis, scaled to a maximum of 50%. (F) IQR NUSE, a metric that indicates the amount of random noise in an expression profile, scaled to a maximum of 0.10. (G-L) Subtype proportions and association with age stratified by dataset. Overall, and within the TCGA samples, age was correlated with the Proneural subtype. The proportions of the subtypes are shown for each dataset and were approximately equal. PN, Proneural; NL, Neural; CL, Classical; MES, Mesenchymal.

Figure S3. Comparison of TCGA with previously published glioma subtypes. (A) Matrix of subtype calls for TCGA core samples (n=173) as predicted by TCGA or Phillips et al. centroids (B) TCGA Core samples clustered with Phillips et al. 35 gene signature. To obtain the best coverage of the Phillips gene signature, gene expression data for 33 of the 35 gene signatures were mapped to the Agilent data platform. (C) Matrix of subtype calls for TCGA core samples (n=173) as predicted by TCGA or Freije et al. gene lists. (D) TCGA Core samples clustered with Freije et al. 44 gene signature. Gene expression data for 40 of the 44 gene signatures were mapped to the Unified dataset. Genes were ordered according to Freije et al. and samples were hierarchically clustered. Samples were color coded based on the ClaNC subtype predictions (Proneural, purple; Neural, green; Classical, blue; Mesenchymal, red).

Figure S4. Genome wide view of copy number alterations in the four major subtypes of glioblastoma. Average copy number was determined for positive and negative copy number estimates separately using 244,000 probes and 150 tumor samples derived from the largest single copy number dataset collected as part of TCGA project (MSKCC 244K Agilent aCGH platform). Normalized copy number was the log₂ ratio to copy number 2. (A) Average copy number calculated for each the four major subtypes of GBM. Genome regions with alterations seen significantly more frequent in one of the subtypes are enlarged. (B) *PDGFRA*, (C) *EGFR*, (D) *CDKN2A/B*, (E) *CDK4*, (F) *NF1*. (G) Average copy number calculated for each of the four major subtypes of GBM as predicted in the Beroukhi et al. dataset (Beroukhi et al., 2007), one of the four datasets in the Validation dataset.

Figure S5, Gene structure of *PDGFRA* indicating missense mutations found in the TCGA sample set.

Figure S6. Variable histopathological features of each GBM subtype (A) Proneural. One tumor (i) contained neoplastic fibrillary and gemistocytic (open arrows) astrocytes, numerous mitoses (arrowhead), widespread endothelial hyperplasia (white arrows), and pseudopalisading necrosis (black arrow). A second tumor (ii) contained predominantly neoplastic gemistocytic astrocytes (open arrows), a lymphocytic infiltrate (black arrows), focal endothelial hyperplasia (not shown), and geographic, infarct-like necrosis (not shown). A third tumor (iii, iv) contained neoplastic fibrillary astrocytes that diffusely infiltrated the neocortex (iii) and featured perineuronal satellitosis (iii, white arrow), focal endothelial hyperplasia (not shown), and no necrosis. Distinct areas of this tumor contained a myxoid background (iv, black arrow). Scale bar (100 μ m) in panel i is applicable for all panels. (B) Neural. One tumor (i, ii) contained neoplastic fibrillary (i, open arrow) and multinucleated (giant cell) astrocytes (ii, white arrows), minigemistocytes (i, black arrows), and oligodendroglia (i, white arrows), the latter of which were characterized by round, regular nuclei with crisp nuclear membranes and a prominent nucleolus. Scattered mitoses (i, arrowhead), perivascular lymphocytes (ii, black arrow), focal endothelial hyperplasia (not shown), and no necrosis were evident in this tumor. A second tumor (iii, iv) contained neoplastic fibrillary (iii) and gemistocytic (iii, black arrow) astrocytes, a prominent lymphocytic infiltrate (iv, black arrow), widespread endothelial hyperplasia (iv, white arrow), and focal geographic, infarct-like necrosis (not shown). A third tumor (v, vi) contained densely (v) and loosely (i) packed neoplastic fibrillary astrocytes and featured scattered mitoses (v, arrowhead), florid endothelial hyperplasia (v, black arrows), and focal pseudopalisading necrosis (not shown). Scale bar in i is applicable to all panels. (C) Classical. One tumor (i, ii) contained neoplastic fibrillary astrocytes (i) and oligodendroglia (ii, open arrows), florid endothelial hyperplasia (i, ii, white arrows), and pseudopalisading necrosis (i, black arrow). A second tumor (iii, iv) contained densely (iii) and loosely (iv) packed neoplastic fibrillary astrocytes and featured abundant mitoses (iii, arrowhead) and both geographic and pseudopalisading necrosis (not shown). A third tumor (v) contained neoplastic fibrillary astrocytes and florid endothelial hyperplasia (white arrows). Scale bar in ii is applicable to panels iii-v. (D) Mesenchymal. One tumor (i, ii) contained predominantly neoplastic gemistocytic astrocytes (i, open arrow) and featured focal endothelial hyperplasia (ii, white arrow), geographic, infarct-like necrosis (ii, black arrow), and focal pseudopalisading necrosis (i, white arrow). A second tumor (iii, iv) contained neoplastic fibrillary astrocytes (iii), a prominent lymphocytic infiltrate (iv, white arrow), widespread endothelial hyperplasia (iv, black arrow), and focal geographic, infarct-like necrosis (not shown). A third tumor (v) contained neoplastic fibrillary and gemistocytic astrocytes, scattered mitoses (v, arrowhead), focal endothelial hyperplasia (not shown), and no necrosis. Scale bar in i is applicable for panels iii-v.

Figure S7. Survival analysis according to GBM subtype. Data for the TCGA dataset and the validation dataset were combined for survival analysis. (A) Glioblastoma samples (B) Glioblastoma and lower grade gliomas.

Table S1. Frequencies and associated p-values from a Fisher Exact two sided tests of copy number alterations and GBM subtype and Chi-squared Tests of Independence. (see excel file).

Table S2. Frequencies and associated Fisher Exact two sided test p-values of mutations and GBM subtype. Only mutations found in at least four samples were tested.

Gene	Proneural	Neural	Classical	Mesenchymal	Total	Chi-square Test of Indep.
	(n=37)	(n=19)	(n=22)	(n=38)	# Mut	
TP53	20 (0.10)	4 (>0.20)	0 (0.04)	12 (>0.20)	36	<0.01
PTEN	6 (>0.20)	4 (>0.20)	5 (>0.20)	12 (>0.20)	27	>0.20
NF1	2 (>0.20)	3 (>0.20)	1 (>0.20)	14 (0.07)	20	0.05
EGFR	6 (>0.20)	5 (>0.20)	7 (>0.20)	2 (>0.20)	20	>0.20
IDH1	11 (<0.01)	1 (>0.20)	0 (>0.20)	0 (>0.20)	12	<0.01
PIK3R1	7 (>0.20)	2 (>0.20)	1 (0.28)	0 (>0.20)	10	>0.20
DST	3 (>0.20)	0 (>0.20)	5 (>0.20)	1 (>0.20)	9	>0.20
RB1	1 (>0.20)	1 (>0.20)	0 (>0.20)	5 (>0.20)	7	>0.20
ERBB2	2 (>0.20)	3 (>0.20)	1 (>0.20)	1 (>0.20)	7	>0.20
EGFRvIII	1 (>0.20)	0 (>0.20)	5 (>0.20)	1 (>0.20)	7	>0.20
PIK3CA	3 (>0.20)	1 (>0.20)	1 (>0.20)	1 (>0.20)	6	>0.20
FKBP9	3 (>0.20)	1 (>0.20)	2 (>0.20)	0 (>0.20)	6	>0.20
SYNE1	3 (>0.20)	0 (>0.20)	1 (>0.20)	1 (>0.20)	5	>0.20
PDGFRA	4 (>0.20)	0 (>0.20)	0 (>0.20)	0 (>0.20)	4	>0.20

Shaded rows correspond to regions with significant chi-squared tests after adjusting for multiple testing.

Numbers between parentheses represent adjusted p-values of the probability that the number of events in a specific subtype is different than that of the remaining samples combined (two sided Fisher's exact test). Bolded entries show significant differences.

Table S3. Gene ontology analysis for genes highly expressed in GBM Subtypes from the ClANC 840 list. (see excel file)

Table S4. Distribution of co-mutations across GBM subtypes.

Gene	Proneural n=37	Neural n=19	Classical n=22	Mesenchymal n=38	Total # Mut
PTEN_TP53	3 (8%)	1 (5%)	0 (0%)	7 (18%)	11
IDH1_TP53	9 (24%)	1 (5%)	0 (0%)	0 (0%)	10
EGFR_TP53	4 (11%)	2 (11%)	0 (0%)	1 (3%)	7
NF1_PTEN	0 (0%)	1 (5%)	0 (0%)	6 (16%)	7
PIK3R1_TP53	5 (14%)	1 (5%)	0 (0%)	0 (0%)	6
NF1_TP53	1 (3%)	0 (0%)	0 (0%)	4 (11%)	5
PTEN_RB1	1 (3%)	1 (5%)	0 (0%)	3 (8%)	5
EGFR_PTEN	2 (5%)	0 (0%)	2 (9%)	0 (0%)	4
RB1_TP53	0 (0%)	1 (5%)	0 (0%)	3 (8%)	4

Table S5. Clinical Analysis statistical tests and p-values

Variable	Test Performed	p-value
Age*	2-way, unbalanced ANOVA (controlling for collection center)	0.04 (TCGA) 0.06 (TCGA—no Duke) 0.07 (Validation)
	1-way ANOVA	0.01 (TCGA) 0.05 (Validation)
	Multinomial generalized linear model	<0.001 (TCGA) 0.01 (Validation)
Karnofsky Score	2-way, unbalanced ANOVA	0.23 (TCGA)
	1-way ANOVA	0.04 (TCGA)
	Multinomial generalized linear model	<0.001 (TCGA)
Sex	Chi-squared test of independence	0.13 (TCGA) 0.50 (Validation)
Survival	Mantel-Haenszel test	0.90 (TCGA) 0.40 (Validation)
<i>MGMT</i> Methylation	Chi-squared test of independence	0.54 (TCGA)
Secondary or Recurrent	Chi-squared test of independence	0.33 (TCGA)
TCGA Batch	Chi-squared test of Independence	0.24 (TCGA)
Collection Center*	Chi-squared test of Independence	0.01 (TCGA)
% Necrosis*	2-way ANOVA, after logit transformation, controlling for collection center	0.01 (TCGA)
% Tumor Nuclei	2-way ANOVA, after logit transformation, controlling for collection center	0.68 (TCGA)

Table S6. Number of bases sequenced per sample per gene. (see excel sheet)

Table S7. Patient Characteristics. (See excel sheet)

Experimental Procedures

Factor Analysis We assume, *per gene*, a standard factor analysis model for the relationship of the standardized gene measurement (y) to the true underlying gene expression (x),

$$y_{ai} = \beta_a x_i + \varepsilon_{ai},$$

$$y_{ei} = \beta_e x_i + \varepsilon_{ei},$$

$$y_{ui} = \beta_u x_i + \varepsilon_{ui},$$

where i indexes a sample, a , e , and u index the Agilent, Exon and U133 platforms, respectively, and ε is the platform specific noise. The x_i are assumed to have a normal distribution with mean zero and standard deviation one; the vector of error terms ε_i is assumed to have a multivariate normal distribution with zero mean and a diagonal covariance matrix, Ψ . The parameters β and Ψ are unknown and are estimated using an EM algorithm within the package `factanal`. With these estimates and a given vector y of observed measurements, estimates of x are a linear combination of the observed y measurements given by,

$$\hat{x} = \hat{\beta}^T (\hat{\beta} \hat{\beta}^T + \hat{\Psi})^{-1} y.$$

Because the inputs y from each platform are standardized values, the estimates $\hat{\beta}$ are exactly equivalent to the correlation across all samples of the input y and the estimate $\hat{x}$.

Rescaling of Unified Expression Measures Each expression platform has a different estimate of its variation between samples. To estimate a single estimate of variation, s , we focus on using information from only those platforms whose gene measurements were well correlated with our unified estimate $\hat{x}$. When possible, we estimated the rescaling factor s as the average MAD value of the platforms that had $\hat{\beta} \geq 0.70$ (possibly only a single platform). This gave an estimate of s for all but 631 genes (and all of

the non-filtered genes used in the estimates of clusters). To obtain an estimate for all 11,861 genes with unified estimates (used in other parts of the analysis) we lowered the cutoff to $\hat{\beta} \geq 0.50$. This gave an estimate of s for all but 158 genes. For these remaining genes, we just took the average MAD across all three platforms.

Gene Filtering. As explained in the main text, three filters were applied to the genes. The first filter kept only genes in which at least two of the platforms had a high correlation with the unified gene expression estimates ($\hat{\beta} \geq 0.70$ for at least two platforms). 9,255 genes passed this criterion. The goal of this filter was not to reduce the analysis to a small number of genes, but rather to eliminate poorly-behaving genes. The second filter eliminated genes that did not have large variability across patients; this was the main filter that was used to reduce the number of genes to a reasonable size for clustering. We filtered based on the individual MAD values per platform. Specifically, we required that the MAD of each platform be greater than 0.5, but only in those platforms that had a high correlation with the unified expression value ($\hat{\beta} \geq 0.50$) and thus had a reasonably high contribution to the unified gene expression estimate. We chose a slightly smaller $\hat{\beta}$ value here because even for genes with $\hat{\beta} \approx 0.50$, the contribution of the platform to unified gene expression value can be as much as 20% (though for genes that pass the first filter, this is not generally a problem). This resulted in 2,120 that satisfied this requirement. Combined with the first filter based on the $\hat{\beta}$ values this gave 1,903 genes. The last filter removed genes with extremely variable estimates of standard deviation. Specifically, we excluded genes if the ratio of the individual MAD and the averaged MAD (or its inverse) was greater than 1.5 (again restricted to platforms with $\hat{\beta} \geq 0.50$). 8,795 genes passed this criteria and in combination with the previous two filters, this resulted in 1,740 genes.

SAM/ROC

For the SAM statistics table, SAM (package siggenes version 1.12.0) is used to identify genes that are differentially expressed in bi-classifications of one subtype vs. all the others. The SAM d statistic is shown in the table for each gene and each subtype.

For the ROC table, the ROC (Receiver Operating Characteristic) curves are drawn as follows. The expression values for each gene are used to order the samples. If a sample belongs to a particular subtype, it is considered true, and false otherwise. The AUC (Area Under the Curve) is then calculated for each gene and each subtype. R package 'ROC' version 1.12.0 is used for the calculations.

Comparison of Subtypes with Previous High-Grade Glioma Classifications. Prior profiling studies have identified subgroups of high grade gliomas (Freije et al., 2004; Phillips et al., 2006). Using the signatures from Phillips et al and Freije et al, subtypes were predicted on the TCGA core samples (n=173). For the Phillips et al subtype prediction, we used the Agilent platform as many of the genes that represented the Proliferation subgroup were not present on the unified data set. We used a single platform, Agilent, to compare the 33/35 gene signature with our four subgroups. One Proneural gene and Mesenchymal gene were not mapped to the Agilent platform. For the Freije et al subtype prediction, 338 genes from the 595 probe sets were identified and used in the unified data set. Subtype predictions on the TCGA data was performed using single sample predictor (Hu et al., 2006). TCGA core samples were clustered with Phillips et al. 35 gene signature or Freije et al. reduced 44 gene signature and color coded according to TCGA four GBM subtypes. High similarity was observed for clusters that identify the Proneural and Mesenchymal subtypes in both the Phillips and Freije gene signatures (Figure S3). Freije et al. HC1B gene set was similar to the Neural subtype. However, neither signature contained a set of genes with high expression limited to the Classical subtype.

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